

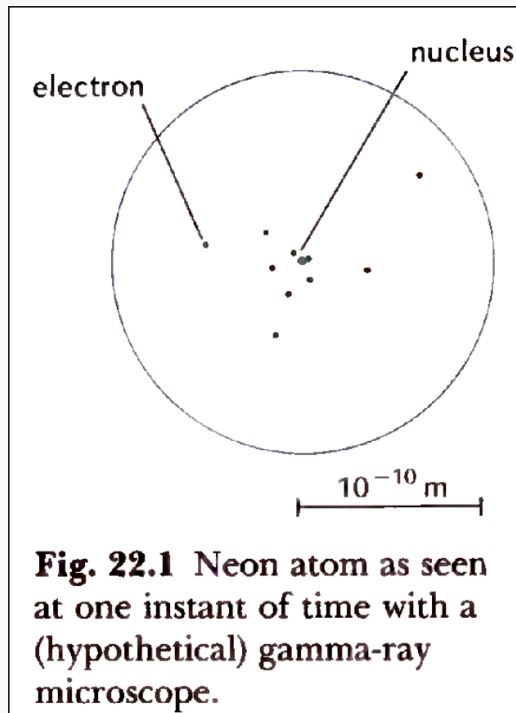
CHP-1 Coulomb's Law

General Physics 2

2026-08-24

Electric Force

- Ordinary matter solids, liquids, gases consists of atoms each with a nucleus and swarms of electrons.



What is the force that holds electrons near the nucleus?

- The force is the electric force between protons in the nucleus and electrons! (might be also called electrostatic force)

So if there is an attractive force between electrons and protons why they don't get on top of each other?

- Its because of quantum effects. If you try to confine electrons in a region very close to protons, according to uncertainty principle they must have some mean square momentum which is larger the more we try to confine them

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

The electric force not only holds electrons inside atoms, but also holds atoms together in molecules or crystals and hold these building blocks together in macroscopic structures.

In the following weeks we will work on

Electrostatic forces : when particles are at rest or moving very slowly

Magnetic forces : when particles are moving uniform velocity

Electromagnetic forces : Combination of magnetic and electrostatic (Also accelerated particles)

Historical Digression: Empirically it was known since ancient times that if amber is rubbed on fur, it acquires the property of attracting light objects such as feathers. This phenomenon was attributed to a new property of matter called “electric charge”. (electron is the Greek name for amber)

The Greek name for amber was $\eta\lambda\epsilon\kappa\tau\rho\nu$ (Electron) and was connected to the Sun God, one of whose titles was Elector or the Awakener. It is discussed by Theophrastus, possibly the first ever mention of the material, and in the 4th century BC. The modern term electron was coined in 1891 by the Irish physicist George Stoney, using the Greek word for amber (and which was then translated as electrum) because of its electrostatic properties and whilst analyzing elementary charge for the first time. The ending -on, common for all subatomic particles, was used in analogy to the word ion.

(1) If we take an electron and proton the electric force between them resembles gravitation that is it goes like inverse square of distance

$$|F_g| = \frac{1}{4\pi G} \frac{m_e m_p}{r^2}, \quad |F_e| \sim \frac{1}{r^2}$$

- how ever F_e much stronger

$$2 \times 10^{39} \times |F_g| = |F_e|$$

(2) Also $|F_g|$ gravitation is always attractive! whereas $|F_e|$ can be repulsive or attractive!

electron & proton $\rightarrow$ attractive,
electron & electron $\rightarrow$ repulsive,
proton & proton $\rightarrow$ repulsive,
neutron & anything $\rightarrow$ Zero.

(3) The magnitude of the forces between electron-electron proton-proton and electron-proton are the same. This implies the force **does not depend** on mass of the particles instead we define

Electric charge : We assign each particle an electric charge which is either $\pm e$ or 0.

For proton $+e$

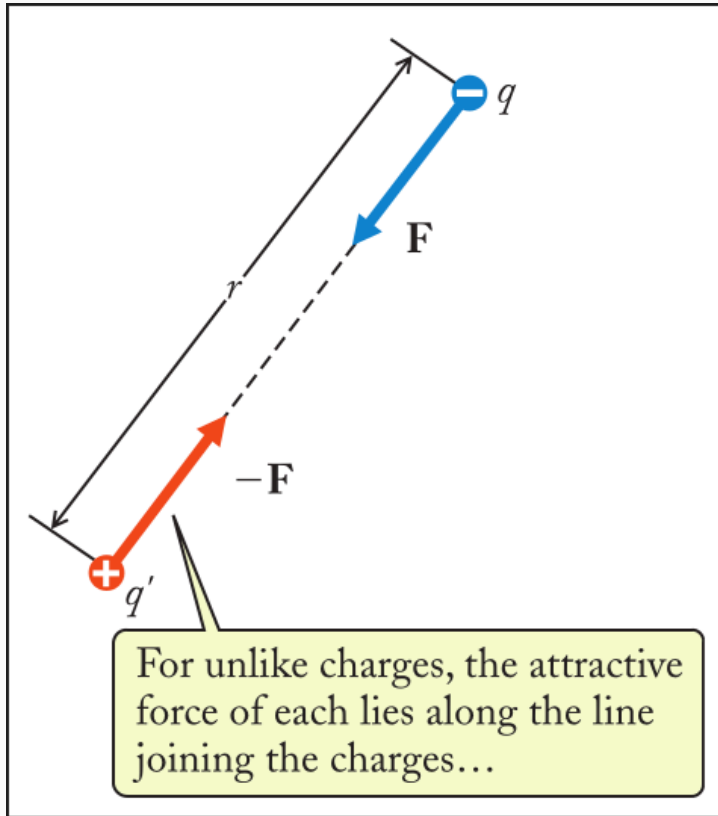
For electron $-e$

For Neutron 0

Thus in terms of electric charge we say

- The force between charges of like sign is repulsive!
- The force between charges of unlike sign is attractive!

Coulomb's Law



The **magnitude** of the electric force that a particle is directly proportional to the product of their charges and inversely proportional to the square of the distance between them.

The **direction** of the force is along the line joining them.

$$|F| = [\text{constant}] \times \frac{qq'}{r^2}$$

$$\vec{F} = [\text{constant}] \times \frac{qq'}{r^2} \hat{r}$$

What about the [constant]? It depends on the system of units, in the SI units, the electric charge of an electron is measured in coulombs

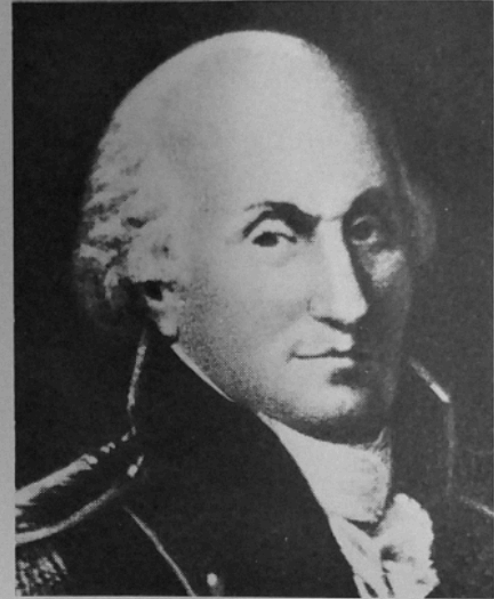
$$e = 1.6 \times 10^{-19} \text{ C}$$

$$[\text{constant}] = 8.99 \times 10^9 \frac{\text{N.m}^2}{\text{C}^2}$$

For historical reasons

$$[\text{constant}] = \frac{1}{4\pi\epsilon_0}; \quad \epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N.m}^2}$$

where ϵ_0 is called permittivity constant.



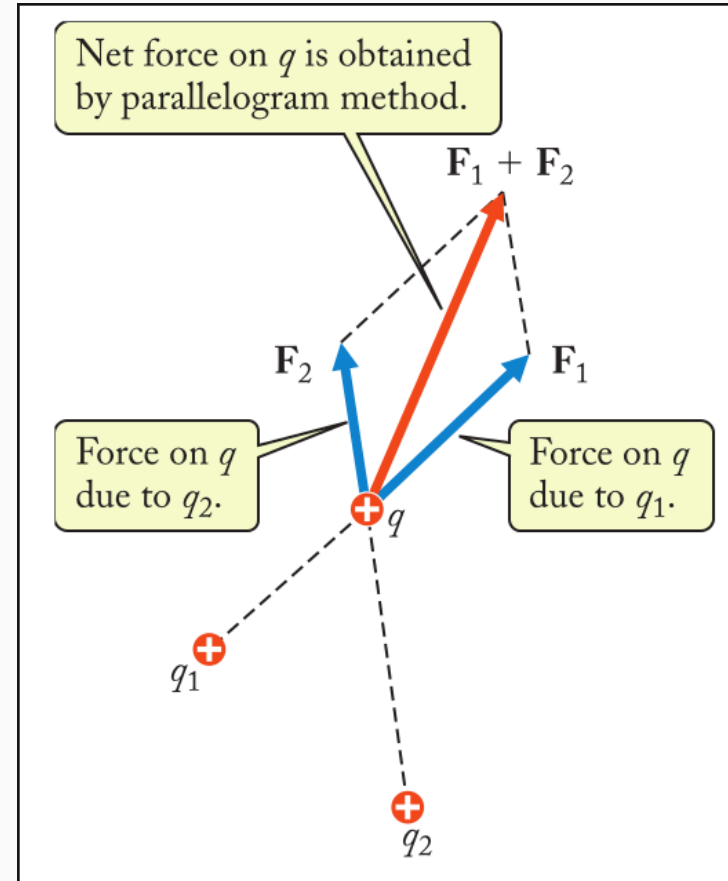
Charles Augustin de Coulomb (koolom), 1736–1806, French physicist. A military engineer by profession, he retired at age 53 from his post as superintendent of waters and fountains so that he could fully pursue scientific research. With the torsion balance, which he invented, he established that the electric force between small charged balls obeys an inverse-square law.

Principle of Superposition

If several point charges $q_1, q_2, q_3, \dots$ simultaneously exert electric forces on the charge q , then the net force on q is obtained by taking the vector sum of the individual forces. Thus, if the vectors representing the individual electric forces produced by $q_1, q_2, q_3, \dots$ are $\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots$ respectively, then the net force is

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \dots$$

this is called **Principle of Superposition**.

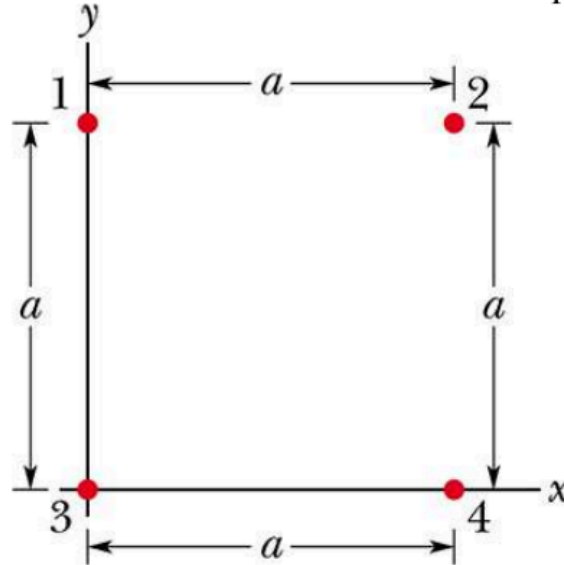


Example: What is the magnitude of the force exerted by the electrons in a cup of water on the protons in a second cup of water at distance of 10m? (take cup of water 250gr with molecular weight of water 18gr. Avogadro's number 6×10^{23} number of particles in 1 mole.)

Example: Point charges Q and $-Q$ are separated by a distance d . A point charge q is equidistant from these charges, at a distance x from their midpoint. What is the force on charge q ?

Example:

In Fig., four particles form a square. The charges are $q_1 = q_4 = Q$ and $q_2 = q_3 = q$. (a) What is Q/q if the net electrostatic force on particles 1 and 4 is zero? (b) Is there any value of q that makes the net electrostatic force on each of the four particles zero? Explain.



Charge Quantization and Charge Conservation

All the particles have charges that are some integer of multiples of the fundamental charge e . Note that quarks have fractional charges!

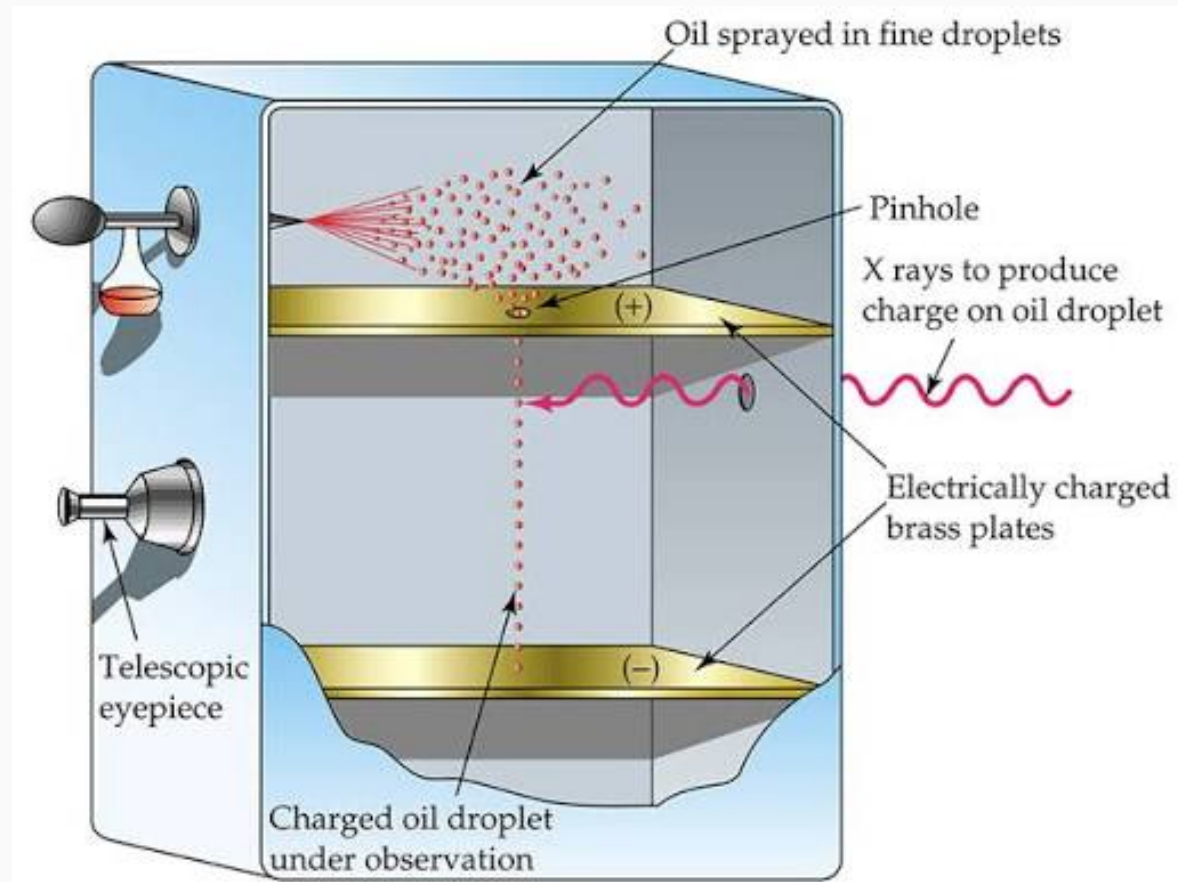
Table 22.3 ELECTRIC CHARGES OF SOME PARTICLES	
Particle	Charge
Photon, γ	0
Neutrino, ν	0
Electron, e	$-e$
Muon, μ	$-e$
Pion, π^0	0
Pion, π^+	$+e$
Pion, π^-	$-e$
Proton, p	$+e$
Neutron, n	0
Delta, Δ^{++}	$+2e$
Delta, Δ^+	$+e$
Delta, Δ^0	0
Delta, Δ^-	$-e$

Since charge exists only in discrete packets we say that charge is **quantized**.

Millikan Experiment

An important experiment in which the charge of small oil droplets was determined was carried out by Millikan. Millikan discovered that the charge on the oil droplets was always a multiple of the charge of the electron (e , the fundamental charge). For example, he observed droplets with a charge equal to $\pm e$, $\pm 2e$, $\pm 3e$, etc., but never droplets with a charge equal to $\pm 1.45e$, $\pm 2.28e$, etc. The experiments strongly suggested that the electric charge, q , is said to be quantized. q is the standard symbol used for charge as a variable.

Millikan experiment setup



The electric charge is a conserved quantity, in any reaction involving charged particles the total charges before and after are always the same.

matter-antimatter annihilation:



$$-e + e \rightarrow 0$$

radioactive disintegration of a neutron:



$$0 \rightarrow e + (-e) + \nu$$

The same argument applies to all macroscopic electric processes such as operation of electric machines, capacitors, thunderbolts; such undoubtedly electric charge is always conserved!

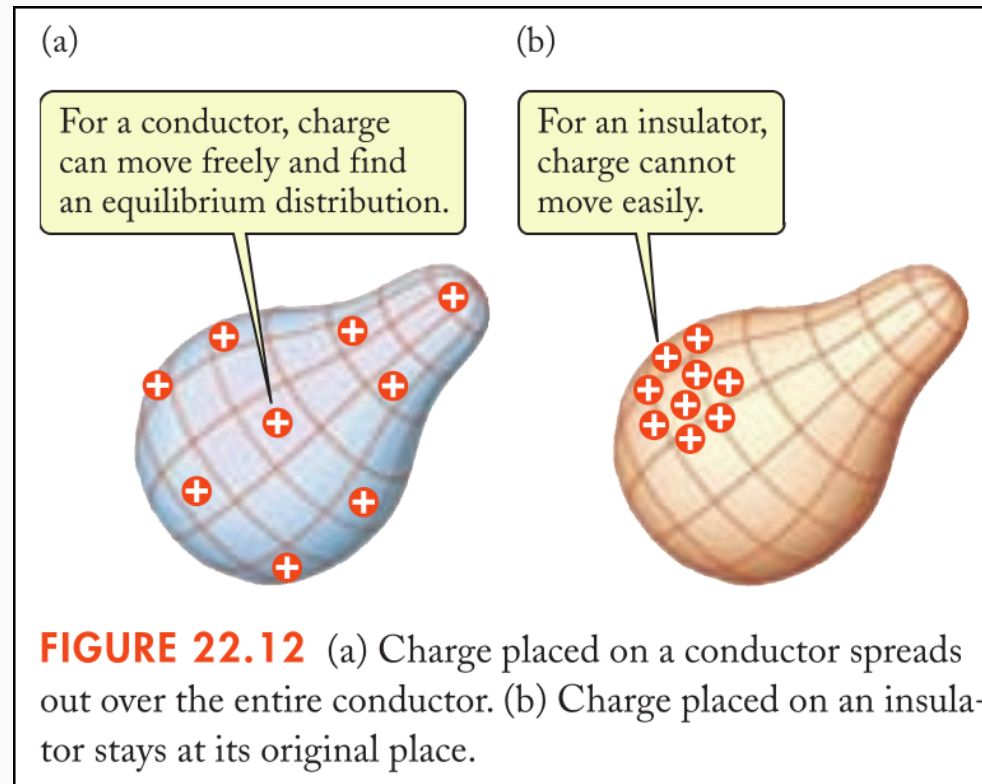
Conductors and Insulators

Conductor : A conductor such as copper, aluminium or iron is a material that permits the motion of electric charge through its volume.

Insulator : Insulator is a material that does not permit motion of electric charge.

- All metals are good conductors; in metals the motion of charge is due to motion of electrons. Some of the electrons are bound to the atoms and some are free.
- Electrons in a metallic conductor are usually said to form free electron gas.
- Liquid containing ions (atoms missing or with excess electrons) are also good conductors, assume e.g. salt water ions of Na^+ and Cl^- . Liquid conductors are also called **electrolytes**.
- Ordinary gases are insulators, but ionised gases are good conductors. Such ionised gas is called **plasma**.

Superconductors are materials that are perfect conductors, allowing charge to move without any hindrance. In these chapters we discuss only conductors and insulators.



Charging by friction or induction

- Electricity was first discovered through friction. The ancient Greeks noticed that rods of amber (elektron, in Greek), when rubbed with a cloth or with fur, gave off sparks and attracted small bits of straw or feathers. Below is a discussion on the sign of the electric charge that is accumulated on the glass rod.

We will end this chapter with a few brief comments on frictional electricity. It is easy to accumulate electric charge on a glass rod merely by rubbing it with a piece of silk. The silk becomes negatively charged and the glass positively charged. The rubbing motion between the surfaces of the silk and the glass rips charges off one of these surfaces and makes them stick to the other, but the detailed mechanism is not well understood. It is believed that what is usually involved is a transfer of ions from one surface to the other. Contaminants residing on the rubbed surfaces play a crucial role in frictional electricity. If glass is rubbed with an absolutely clean piece of silk or other textile material, the glass becomes negatively charged rather than positively charged — ordinary pieces of silk apparently have such a large amount of dirt on their surfaces that the charging process is dominated by the dirt rather than by the silk. Even air can act as contaminant for some surfaces; for instance, careful experiments on the rubbing of platinum with silk show that in vacuum the platinum becomes negatively charged, but that in air it becomes positively charged.

The electric charge that can be accumulated on the surface of a body of ordinary size (a centimeter or more) by rubbing may be as much as 10^{-9} to 10^{-8} coulomb per square centimeter. If the charge

Tribo-Electricity and Friction.

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Nottingham.

(Communicated by Sir William Hardy, F.R.S.—Received February 4, 1926.)

I. Glass and Textiles.

This paper deals with the changes in the condition of glass surfaces when rubbed with flexible materials, chiefly cotton, linen, and silk, of varying purity. The condition is ascertained by two different, but associated, properties :—

- (a) The electric charges displayed by the glass surfaces when rubbed on textiles or on one another.
- (b) The coefficient of friction found between two of them after being subjected to identical rubbing.

We have derived great help in the friction work both as regards theory and technique from the papers on “Boundary Lubrication,” by Sir William Hardy and his colleague (1, 2, 3 4).

A solid surface may be modified in two distinct ways :—

(1) *By Addition of Material.*—A solid surface exposed to an atmosphere containing water (or other) vapours, condenses and retains these; or if solid or liquid matter be rubbed on the surface, this will in general acquire and retain a covering film. The actual exposed surface is thus a complex of various materials, so that tribo-electric and frictional effects are complicated. We know that the

The following definite effects have been found :—

(1) A glass surface starting in a standard state of purity adsorbs condensable material if left in the air and its coefficient of friction (μ) *slowly* descends in several days from some high value, say 1.2 to about 0.7.

(2) Ordinary textile materials when rubbed on the standard glass bring about the lowering of μ *quickly*. Continued rubbing will bring down even as low as 0.13. For a value of $\mu = 0.18$ the charge on the glass due to rubbing becomes $+^{\text{ve}}$, having been $-^{\text{ve}}$ for higher values of μ .

(3) Textile materials, rendered very clean by prolonged treatment with solvents (benzene, etc.), slowly reduce μ for the glass on which they are rubbed to about 0.6, as the limit. The friction never descends to 0.18 and the glass remains $-^{\text{ve}}$ and never acquires a $+^{\text{ve}}$ charge.

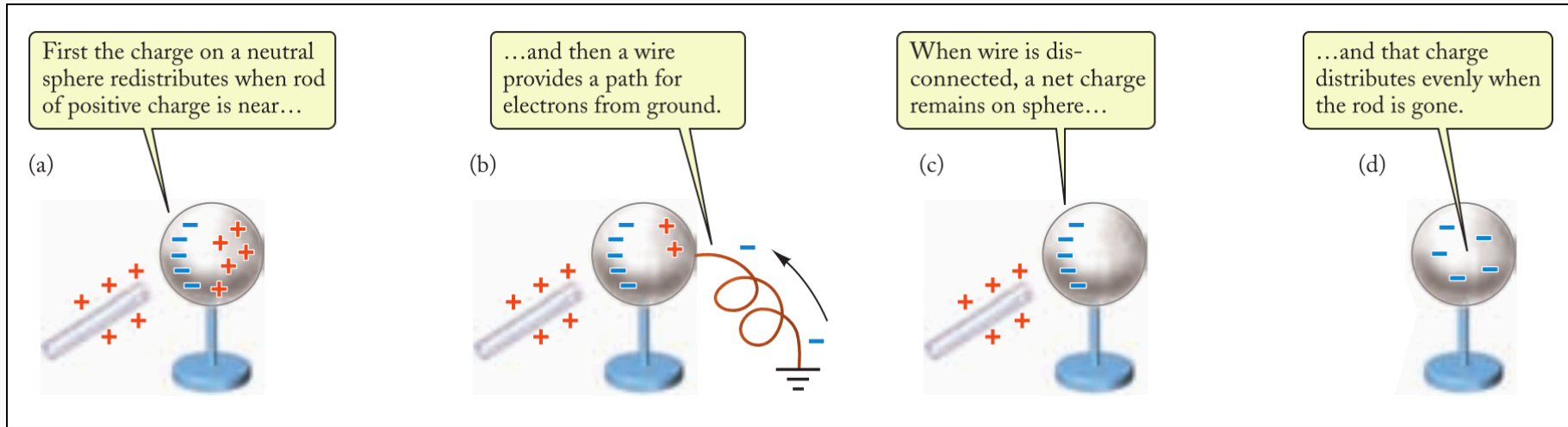
(4) When a glass surface has attained a very low value of μ , say 0.13, it acquires a higher value (even 0.54) when heated, or placed in a vacuum, or left in air.

(5) When glass has medium values of μ , the sign of the charge due to rubbing can be made $+^{\text{ve}}$ or $-^{\text{ve}}$ according as the rubbing is gentle or violent, and when the yarns and fabrics are used, according as the rubbing is along or across the fibre.

(6) Two standard glass surfaces acquire no charges when rubbed together, but if one has been rubbed by a textile more than the other it is $+^{\text{ve}}$ to the other.

(7) If one of two identical surfaces of glass be heated and then rubbed on the other, it becomes $-^{\text{ve}}$ to it.

- A second, quite different way to charge an object is called charging by induction. Once we have accumulated some charge, say, positive charge on a rod of glass, we can produce charges on other bodies by the induction process.



Photocopy machine

